# IonQ Qollab Hackathon Mathematician Documentation

Team Briskit

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A central component of this project is representing optimization problems using cost functions suitable for QAOA. This mathematics documentation establishes key properties of these formulations, develops the classical algorithm baseline for tree restoration, and investigates some limitations of these related methods.

## 1 Preliminary Notation

We use the following notation, where we have two identical crews and $n$ repairs. We say that repair $i$ takes $p_i > 0$ hours with weight $w_i \geq 0$. Define $r_i = \frac{w_i}{p_i}$ and:

$$s_i = \begin{cases} 1, & \text{crew A} \\ -1, & \text{crew B.} \end{cases}$$

Each crew processes its repairs in nonincreasing order of $r_i$, which serves as the priority. All repairs are independent. Our objective is to minimize the function:

$$F(s) = \sum_i w_i C_i(s),$$

where $C_i(s)$ denotes the time when repair $i$ finishes, measured from when the crew begins working. We show a series of claims.

## 2 Two-Crew Cost and Sum-of-Squares Decomposition

**Claim:**

$$F(s) = \text{constant } + \frac{1}{2} \sum_{i<j} J_{ij} s_i s_j \qquad J_{ij} = p_i p_j \min(r_i, r_j)$$

*Proof:* Take any pair of repairs $i, j$ on the same crew. Notice that every repair contributes $w_i p_i$. Now, if $r_i > r_j$, by the ordering, repair $i$ must come first. This adds a factor of:

$$(w_i p_i + w_j(p_i + p_j)) - w_i p_i - w_j p_j = p_i w_j = p_i p_j r_j$$

to the objective. Similarly, whenever $r_j > r_i$, this extra contribution is $p_j w_i = p_i p_j r_i$. In the event of a tie, these values are the same.
We can then define an interaction coefficient in the form of a matrix:

$$J_{ij} = p_i p_j \min(r_i, r_j).$$

Our cost function can then be simplified to the following, by taking out the baseline costs:

$$\begin{aligned} F(s) &= \sum_i w_i p_i + \sum_{i<j} J_{ij} \mathbf{1}_{\{s_i = s_j\}} \\ &= \sum_i w_i p_i + \sum_{i<j} J_{ij} \frac{1 + s_i s_j}{2} \\ &= \left[ \sum_i w_i p_i + \frac{1}{2} \sum_{i<j} J_{ij} \right] + \frac{1}{2} \sum_{i<j} J_{ij} s_i s_j \end{aligned}$$

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**Claim:** Let the $L$ distinct ordered positive ratios be:

$$\rho_1 > \rho_2 > \cdots > \rho_L > 0, \qquad \rho_{L+1} = 0,$$

and define $d_\ell = \rho_\ell - \rho_{\ell+1}$. Then,

$$F(s) = \text{constant } + \frac{1}{4} \sum_{\ell=1}^{L} d_\ell \left( \sum_{i : r_i \geq \rho_\ell} p_i s_i \right)^2.$$

*Proof:* Notice that, by basic telescoping:

$$\min(r_i, r_j) = \sum_{\ell=1}^{L} d_\ell \mathbf{1}_{\{r_i \geq \rho_\ell\}} \mathbf{1}_{\{r_j \geq \rho_\ell\}}$$

Using the $n \times n$ matrix $J$ before, we can observe that:

$$\begin{aligned} J_{ij} &= p_i p_j \min(r_i, r_j) \\ &= \sum_{\ell=1}^{L} d_\ell (u_\ell)_i (u_\ell)_j \\ \implies J &= \sum_{\ell=1}^{L} d_\ell u_\ell u_\ell^T \end{aligned}$$

where $u_\ell$ for each level $\ell$ is the $n$-dimensional column vector with:

$$(u_\ell)_i = \begin{cases} p_i, & r_i \geq \rho_\ell \\ 0, & r_i < \rho_\ell. \end{cases}$$

It is then clear that:

$$\begin{aligned} F(s) &= \left[ \sum_i w_i p_i + \frac{1}{2} \sum_{i<j} J_{ij} \right] + \frac{1}{2} \sum_{i<j} J_{ij} s_i s_j \\ &= \frac{3}{4} \sum_i w_i p_i + \frac{1}{2} \sum_{i<j} J_{ij} + \frac{1}{4} s^T J s \\ &= \left[ \frac{3}{4} \sum_i w_i p_i + \frac{1}{2} \sum_{i<j} J_{ij} \right] + \frac{1}{4} \sum_{\ell=1}^{L} d_\ell \left( \sum_{i: r_i \geq \rho_\ell} p_i s_i \right)^2 \end{aligned}$$

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**Claim:** The rank of $J$ is $L$.
*Proof:* Note that the $j$-th column of $J$ is:

$$\sum_{\ell=1}^{L} d_\ell (u_\ell)_j u_\ell.$$

From this it is clear that the rank is at most $L$ since each column is a linear combination of the $L$ vectors $u_1, \ldots, u_L$. We can observe that these vectors are linearly independent from looking at the lowest positive urgency level.

$$\begin{aligned} r_{i_k} = \rho_k, \qquad k = 1, \ldots, L, \\ \sum_{\ell=1}^{L} \alpha_\ell u_\ell = 0 \implies \sum_{\ell=k}^{L} \alpha_\ell = 0 \quad (k = 1, \ldots, L) \\ \implies \alpha_1 = \cdots = \alpha_L = 0. \end{aligned}$$

Now, take any vector $z \in \ker J$. We have:

$$0 = z^T J z = \sum_{\ell=1}^{L} d_\ell (u_\ell^T z)^2$$

which implies that $u_\ell^T z = 0$ for all $\ell$. Thus, every vector in $\ker J$ is orthogonal to the $L$ independent vectors, and hence $\dim \ker J \leq n - L$. The rank-nullity theorem finishes. ■

# 3 Exact Tree Restoration via Dynamic Programming

We consider a rooted tree with $n$ damaged lines, with line $j$ having weight $w_j \geq 0$. We want to restore exactly $k$ lines, $1 \leq k \leq n$. A line counts if and

only if it and every line between it and the source are restored. We show an algorithm to find the $k$ lines with largest total weight with running complexity $O(nk)$. We call a line energized when it and all its damaged ancestors are repaired.

First, we explain a reduction algorithm. For any repair set, we remove the repaired lines that are not energized. Then, the remaining set must contain every damaged ancestor of each selected line. If there are fewer than $k$ lines, we can repeatedly add an unselected line closest to the source, which cannot decrease value. This shows that there exists some optimal configuration that consists of exactly $k$ lines containing all their damaged ancestors.

This reduction reduces the problem to a tree knapsack problem, with capacity $k$. Kumabe, Maehara, and Sin'ya [1, Example 2] give an $O(nk)$ algorithm for this problem. We then explain an explicit recurrence.

We use a depth-first preorder to list the lines as $v_1, v_2, \ldots, v_n$. We let $a(i)$ be the first position after the subtree of $v_i$. We define $F(i,t)$ as the maximum weight of selecting exactly $t$ lines in positions $i, \ldots, n$, and we require every selected line's ancestors within that suffix to be selected. Then, notice that:

$$F(i,t) = \max\{F(a(i),t), w_{v_i} + F(i+1,t-1)\}, \qquad 1 \le t \le k.$$

The first option skips $v_i$ and its descendants, while the second repairs $v_i$. Now, $F(i,0) = 0$ and $F(n+1,t) = -\infty$ for $t > 0$. We can then compute the rows backward. There are $O(n(k+1))$ states with constant work per state, and thus the time is $O(nk)$.

We now give an example of this implementation with 6 lines $A, B, C, D, E, F$. The source connects to $A$, $A$ has children $B$ and $C$, $B$ has children $D$ and $E$, and $C$ has child $F$. Let the weights be $1, 4, 2, 8, 5, 7$, respectively, and $k = 4$. The preorder is $(A, B, D, E, C, F)$, and the jump indices are $(7, 5, 4, 5, 7, 7)$. Skipping at $B$ gives $F(5,3) = -\infty$ while selecting gives $4 + F(3,2) = 17$. Then, $F(1,4) = 1 + F(2,3) = 18$, achieved by repairing $\{A, B, D, E\}$. Then, all four lines are energized and their total weight is $1 + 4 + 8 + 5 = 18$.

This algorithm provides the classical benchmark, so that we can measure the quality of approximate solutions against the optimum.

# 4 Limits of Level-by-Level Optimization

We first present a level-by-level greedy method, by minimizing the current imbalance without considering anything in the future. The goal is to divide repairs between the two crews so that important repairs finish ASAP. We measure this by minimizing $F = \sum_i w_i C_i$. We start with a signed hour difference of $b = 0$,

and at level $\ell$ we greedily assign that level's repairs to minimize:

$$\left| b + \sum_{i:r_i=\rho_\ell} p_i s_i \right|,$$
$$b \leftarrow b + \sum_{i:r_i=\rho_\ell} p_i s_i.$$

We first show that even with two levels, this is not optimal. Take the three repairs with $p = (1, 1, 2), w = (3, 3, 4), r = (3, 3, 2)$. The greedy method gives:

$$F(g) = 3(1) + 3(1) + 4(3) = 18.$$

However, we can assign the first two to $A$ and repair 3 to $B$, giving:

$$F_{opt} = 3(1) + 3(2) + 4(2) = 17.$$

We now proceed to bound the excess cost. Define $M_\ell = \max_{i:r_i \geq \rho_\ell} p_i$ as the maximum individual repair duration among the first $\ell$ levels.

$$D_0(s) = 0, \qquad D_\ell(s) = \sum_{i:r_i \geq \rho_\ell} p_i s_i,$$
$$s^* \in \operatorname*{arg\,min}_{s \in \{-1,1\}^n} F(s), \qquad F_{opt} = F(s^*).$$

**Claim:**

$$0 \leq F(g) - F_{opt} \leq \frac{1}{4} \sum_{\ell=2}^{L} d_\ell M_\ell^2$$

*Proof:* For any such difference $b$, adding a repair of duration $p$ to the less-loaded crew produces a magnitude of:

$$||b| - p| \leq \max(|b|, p).$$

The level optimizer does at least as well as this split, giving $|D_\ell(g)| \leq M_\ell$, and we clearly have $D_1(g)^2 \leq D_1(s^*)^2$.

$$|D_\ell(g)| \leq \max\left\{ |D_{\ell-1}(g)|, \max_{i:r_i=\rho_\ell} p_i \right\} \leq M_\ell.$$

Therefore,

$$\begin{aligned} 4(F(g) - F_{opt}) &= \sum_{\ell=1}^{L} d_\ell (D_\ell(g)^2 - D_\ell(s^*)^2) \\ &\leq \sum_{\ell=2}^{L} d_\ell D_\ell(g)^2 \leq \sum_{\ell=2}^{L} d_\ell M_\ell^2. \end{aligned}$$

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For integer hours, we now present an exact method for any number of levels whose running time is polynomial in $n$ and the total hours. Let $T$ be the total number of hours, and let $A_i(z)$ be the minimum penalty from completed urgency levels among the first $i$ repairs with difference $z$.

$$p_i \in \mathbb{Z}_{>0}, \qquad T = \sum_{i=1}^{n} p_i,$$
$$r_1 \geq r_2 \geq \cdots \geq r_n \geq 0, \qquad r_{n+1} = 0.$$

Letting $A_0(0) = 0$ and $A_0(z) = +\infty$ for $z \neq 0$, we can compute:

$$A_i(z) = (r_i - r_{i+1})z^2 + \min\{A_{i-1}(z - p_i), A_{i-1}(z + p_i)\}, \qquad z \in \mathbb{Z}.$$

This achieves a final optimum of

$$F_{opt} = \frac{3}{4}\sum_i w_i p_i + \frac{1}{2}\sum_{i<j} J_{ij} + \frac{1}{4}\min_{z\in\mathbb{Z},\ |z|\leq T} A_n(z)$$

# 5 Approximation Guarantee for Urgency-Level Rounding

Choose some $\epsilon > 0$ and let $q = 1 + \epsilon$. Let $a = \min_{i:r_i>0} r_i, b = \max_i r_i > 0$. We round each positive ratio upward to the smallest grid value $aq^m$ for $m = 0, 1, 2, \ldots$ that is at least $r_i$. Denote this rounded value by $\tilde{r}_i$.

$$\tilde{r}_i = 0 \qquad \text{if } r_i = 0.$$

**Claim:** The best plan of the rounded problem is within a factor $1 + \epsilon$ of the optimum.

*Proof:* The rounded instance has the same durations and rounded weights with $\tilde{w}_i = p_i\tilde{r}_i$. Therefore,

$$r_i \leq \tilde{r}_i \leq qr_i \qquad w_i \leq \tilde{w}_i \leq qw_i.$$

Defining the rounded cost as $\tilde{F}(s) = \sum_i \tilde{w}_i C_i(s)$, the above inequalities immediately give that:

$$F(s) \leq \tilde{F}(s) \leq (1 + \epsilon)F(s)$$

as all completion times are nonnegative. Then, letting $s^*$ minimize the true cost and $\hat{s}$ minimize the rounded cost, we have:

$$F(\hat{s}) \leq \tilde{F}(\hat{s}) \leq \tilde{F}(s^*) \leq (1 + \epsilon)F_{opt}$$

where $F_{opt} = F(s^*)$. ■

We now derive a sharper guarantee. Define

$$X(s) = \sum_{i<j:s_i\neq s_j} p_i p_j \min(r_i, r_j),$$
$$\tilde{X}(s) = \sum_{i<j:s_i\neq s_j} p_i p_j \min(\tilde{r}_i, \tilde{r}_j).$$

Minimizing the cost is equivalent to maximizing $X$. Therefore,

$$X(s^*) \le \tilde{X}(s^*) \le \tilde{X}(\hat{s}) \le qX(\hat{s}),$$
$$F(s) - X(s) = \frac{1}{2}\sum_i w_i p_i + \frac{1}{2}s^T J s \ge 0.$$

$$\begin{aligned} F(\hat{s}) - F(s^*) &= X(s^*) - X(\hat{s}) \\ &\le \frac{\epsilon}{1+\epsilon} X(s^*) \le \frac{\epsilon}{1+\epsilon} F(s^*) \end{aligned}$$

and so

$$F(\hat{s}) \le \left(1 + \frac{\epsilon}{1+\epsilon}\right) F(s^*).$$

# 6 Interaction Orders and Quadratic Representability

Now note that any function of $n$ yes or no decisions has a unique expansion over subsets $S$ of the decisions, with $s_i$ being $\pm 1$:

$$f(s) = \sum_{S \subseteq \{1,2,\ldots,n\}} c_S \prod_{i \in S} s_i.$$

The share at order $d$ is the sum of $c_S^2$ over subsets of size $d$, divided by the sum over all non-empty subsets.

**Claim:** Every odd order in two-crew assignment vanishes on every network.

*Proof:* Changing the crew names changes $s$ to $-s$ without changing any cost, and so $f(-s) = f(s)$. Notice:

$$\prod_{i \in S}(-s_i) = (-1)^{|S|} \prod_{i \in S} s_i.$$

Then, by uniqueness, $c_S = (-1)^{|S|} c_S$ and so $c_S = 0$ whenever $|S|$ is odd. ■

We now proceed to find exact shares for budgeted restoration. Let $P_j$ contain line $j$ and every upstream damaged line that is required to serve that weight. Now let $h_j = |P_j|$ and $a_{jk} = |P_j \cap P_k|$. In other words, $h_j$ is the path's repair depth and $a_{jk}$ is the number of repair decisions shared by the two paths. Now, set $x_i = \frac{1+s_i}{2} = \mathbf{1}_{\{\text{line } i \text{ is repaired}\}}$. Then,

$$V(s) = \sum_j \frac{w_j}{2^{h_j}} \prod_{i \in P_j} (1 + s_i) \qquad c_S = \sum_{j : S \subseteq P_j} \frac{w_j}{2^{h_j}}.$$

Naturally, now let $E_d = \sum_{|S|=d} c_S^2$. Then,

$$E_d = \sum_{j,k=1}^{n} \frac{w_j w_k}{2^{h_j + h_k}} \binom{a_{jk}}{d}, \qquad \text{share}_d = \frac{E_d}{\sum_{q=1}^{n} E_q} \quad \left(\sum_{q=1}^{n} E_q > 0\right)$$

through a simple calculation, and from here the share is immediate to compute.

# 7 Trapped-Ion Interaction Phase Derivation

Taking $\hbar = 1$, the supplied Hamiltonian is:

$$H(t) = \sum_i g_i(t) X_i (a e^{-i\delta t} + a^\dagger e^{i\delta t}),$$
$$g_i(t) = g_i \mathbf{1}_{\{0 \le t < T_i\}}, \qquad g_i \in \mathbb{R},$$
$$\delta \in \mathbb{R} \setminus \{0\}, \qquad P = \frac{2\pi}{|\delta|}, \qquad [a, a^\dagger] = 1.$$

Here, $X_i$ is the Pauli $X$ on ion $i$, and $a$ and $a^\dagger$ are the usual annihilation and creation operators.
Set $G = \sum_i g_i X_i$ and $D(A) = \exp(Aa^\dagger - A^\dagger a)$.

$$T_i = T = mP, \quad m \in \mathbb{Z}_{\ge 0}, \quad 0 \le t \le T, \quad U(0) = I.$$

Then, using $i\dot{U} = HU$, we have:

$$U(t) = D\left(\frac{1 - e^{i\delta t}}{\delta} G\right) \exp\left(i \frac{\delta t - \sin \delta t}{\delta^2} G^2\right).$$

When $t = T$, $e^{i\delta T} = 1$ and so:

$$U(T) = \exp\left(i \frac{T}{\delta} G^2\right)$$

which gives us that:

$$G^2 = \left(\sum_i g_i^2\right) I + 2 \sum_{i<j} g_i g_j X_i X_j.$$

Therefore, up to a global phase, we have:

$$U(T) = \exp\left(i \sum_{i<j} \frac{2 g_i g_j T}{\delta} X_i X_j\right)$$

in which the vibration returns to the initial state. Now, when the switch-off times are different whole numbers of periods, the resulting spin phases commute and thus add. The ions $i, j$ are then simultaneously active for total time $\min(T_i, T_j)$, giving us:

$$\theta_{i,j} = \frac{2 g_i g_j}{\delta} \min(T_i, T_j), \qquad T_i = m_i P, \quad m_i \in \mathbb{Z}_{\ge 0}.$$

# References

[1] S. Kumabe, T. Maehara, and R. Sin'ya, "Linear Pseudo-Polynomial Factor Algorithm for Automaton Constrained Tree Knapsack Problem," arXiv:1807.04942, 2018. https://arxiv.org/pdf/1807.04942

[2] Y. Tan et al., "Scheduling Post-Disaster Repairs in Electricity Distribution Networks," arXiv:1702.08382, 2017. https://arxiv.org/abs/1702.08382